Letter to the editors

Response to Reviewers — Submission 40

AutoGraphForge: Towards Automated Graph Theory Discovery

Dear Editors,

We thank both reviewers for their careful reading. Below is a complete account of the changes.

A. Changes responding to Reviewer 1

A1 — Grammar (comment 3)

> "counterexample search algorithms **is** built upon"
> → "counterexample search algorithms **are** built upon"

A2 — Grammar (comment 4)

> "the proving stage is **implemented in and** sanity-checked but not yet properly evaluated"
> → "the proving stage is **implemented and** sanity-checked but not yet properly evaluated"

A3 — Sophie necessary conditions, readability (comment 5)

The long clause chain was replaced by short sentences leading with the mechanism:

> "Sophie *necessary conditions* are handled by contraposition. Such a condition has the form $A \Rightarrow \neg C$, where A is a numeric predicate and C a graph class. It is flagged known whenever the contrapositive $C \Rightarrow \neg A$ appears in the table. For instance $(\chi > 2) \Rightarrow \neg\mathsf{bipartite}$ is recognised as the contrapositive of the known bound $\mathsf{bipartite} \Rightarrow \chi \leq 2$."

A4 — Qualitative summary of the inspected conjectures (comment 2) — **ADDED**

A 35-line commented-out draft (a hand-classification of the top 50: 22/12/6/5/5) was **removed** and replaced with a live paragraph and a new table:

Category	Count
Rediscovery recognised by the novelty filter	49
Subsumed by a stronger survivor	2
Decorative hypothesis (class does no work)	1
Promising, universal	28
Promising, class-conditioned	20
Total	**100**

The categories are now computed by `tools/audit_sample.py` from the pipeline's own novelty table, subsumption lattice and support statistics rather than assigned by hand, so the classification is reproducible from the released artefacts.

We report no "probably false" category and say so explicitly: every item survived refutation against the full dataset, so we have no evidence of falsity for any of them, and asserting one would be an unsupported guess.

B. Changes responding to Reviewer 2

B1 — SAT/CP-SAT for the NP-hard invariants — **ADDED**

Absent from the submitted version; added after the CPU-budget accounting:

> "Since the precomputation dominates, and within it the NP-hard invariants dominate, replacing the ILP formulations with a SAT or CP-SAT encoding is the single most promising engineering change available to us. Independence, domination and zero-forcing all admit natural propositional encodings, and modern conflict-driven and local-search solvers are frequently far faster than ILP on instances of this size. A cheaper battery is not merely a saving: it would let us extend the refutation tiers to larger orders, and so reach invariants that are uninformative on the small graphs the present pool is concentrated on. We are grateful to a reviewer for this suggestion."

C. small corrections

C1 — Survivor count (4 sites: abstract, Table 5 caption, §4.2, §6)

I did a mistake when parsing the output of the computations. This number correction does not have any consequence apart from itself.

> "$7{,}935$ surviving conjectures" → "$6{,}522$ surviving conjectures"

C2 — Survivor split (§4.2)

The stated split summed to the old total:

> "split into $3{,}386$ class-conditioned inequalities and $4{,}549$ Sophie **sufficient-**conditions"
> → "split into $2{,}677$ class-conditioned inequalities and $3{,}845$ Sophie conditions"

Recomputed from C1; $2{,}677 + 3{,}845 = 6{,}522$.

C3 — Merge description (Table 5 caption)

The whole reduction was attributed to deduplication, which accounts for only 346 of it:

> "the five sum to the $8{,}281$ raw survivors that the merge step **deduplicates to** $7{,}935$."
> → "the five sum to the $8{,}281$ raw survivors. The merge step reduces these to $6{,}522$ **in two stages:** $1{,}413$ **are refuted by witnesses found in another partition** — each partition grows its own hard seed and never sees the others', so a graph that kills a candidate may sit in the wrong shard — **and a further** 346 **are cross-partition duplicates**."

$8{,}281 - 1{,}413 - 346 = 6{,}522$.

C4 — Ranks in §5

The touch heuristic previously ignored the class hypothesis, so different hypotheses over the same body received identical touch counts:

> "the bipartite class (rank 99 of $7{,}935$ by touch, touch $2{,}053$) and the cubic/regular classes (cubic: rank 72, touch $2{,}181$; regular: rank 100, touch $2{,}053$)"
> → "the bipartite class (rank 91 by touch, touch 922) and the regular class (rank 177, touch 589). Ranks are among the $2{,}677$ survivors that carry a recomputed, hypothesis-aware touch count; **the cubic specialisation (rank**

> 453**, touch** 233**) is now flagged as subsumed by the regular form**, since cubic $\subset$ regular"

C5 — Boolean count (§3.3.1)

The paragraph said 14 booleans and then 16, without explanation:

> "while the 16 booleans act as Sophie hypotheses and as class conditions"
> → "while the 14 booleans — **together with two derived order thresholds (**$n \geq 3$ **and** $n \geq 4$**) that the battery adds precisely so that bounds failing only on the smallest graphs become true class-restricted statements** — give the 16 predicates available to Sophie as hypotheses and class conditions"

C6 — Round-one expansion (§5.1)

> "a ~ 20–28% expansion of T" → "a ~ 17–28% expansion of T"

Per partition against $T = 2{,}860$: 27.7, 17.4, 17.3, 19.7, 20.3 %. The smallest partition adds 494 witnesses, i.e. 17.3 %, so the lower end was overstated.

D. Other changes

D1 — Acknowledgments

> Added: "I also thank anonymous reviewers for helpful suggestions and comments."

D2 — Code availability (addresses Reviewer 1, comment 1)

> Added: "The complete list of surviving conjectures is published there as `results/survivors_by_touch.txt`: all $6{,}522$ survivors in human-readable form, ranked by touch number with the qualitative verdict of Table 2 attached to each, so the open conjectures discussed here can be inspected and picked up directly."

The required edits required me increasing the number of pages slightly. I apologize for this and hope this does not cause any problems.

Yours sincerely,

Jan Pastorek